

Sustainability and Society

Sustainability :

i. Ability to exist and develop for the future generation without depleting the natural resources.

ii. It can be maintained over time.

" Enjoy the present, Save for the future "

planned / organized community :

i. Bottom-up approach : Development Strategies, environmental initiatives, economic growth are driven by the local communities, grassroot Organizations and people rather than dominated or dictated by the

central bodies (which would be
Top-down approach)

- ii. The people who live in an ecosystem understand its challenge better than Outside people.
- iii. planned community is only possible if the people are conscious about the resources.
- iv. Access to education, healthcare and economy by the poor peoples makes the community.

Sustainable community :

- i. Sustainable community is a neighbourhood town, city designed, built and operated in a way that all

residents live a comfy life without depleting the natural resources much.

- ii. Sustainable community is not just one type. It depends on needs, access, environment and people. It differs from one community to others.
- iii. Sustainable community is the ultimate real-world application of Bottom-up approach.

The Brundtland Report :

Official name: Our Common Future

Release date: October 1987.

organization: United Nation World

Commission on Environment and Development
(WCED)

Chair person: Gro Harlem Brundtland

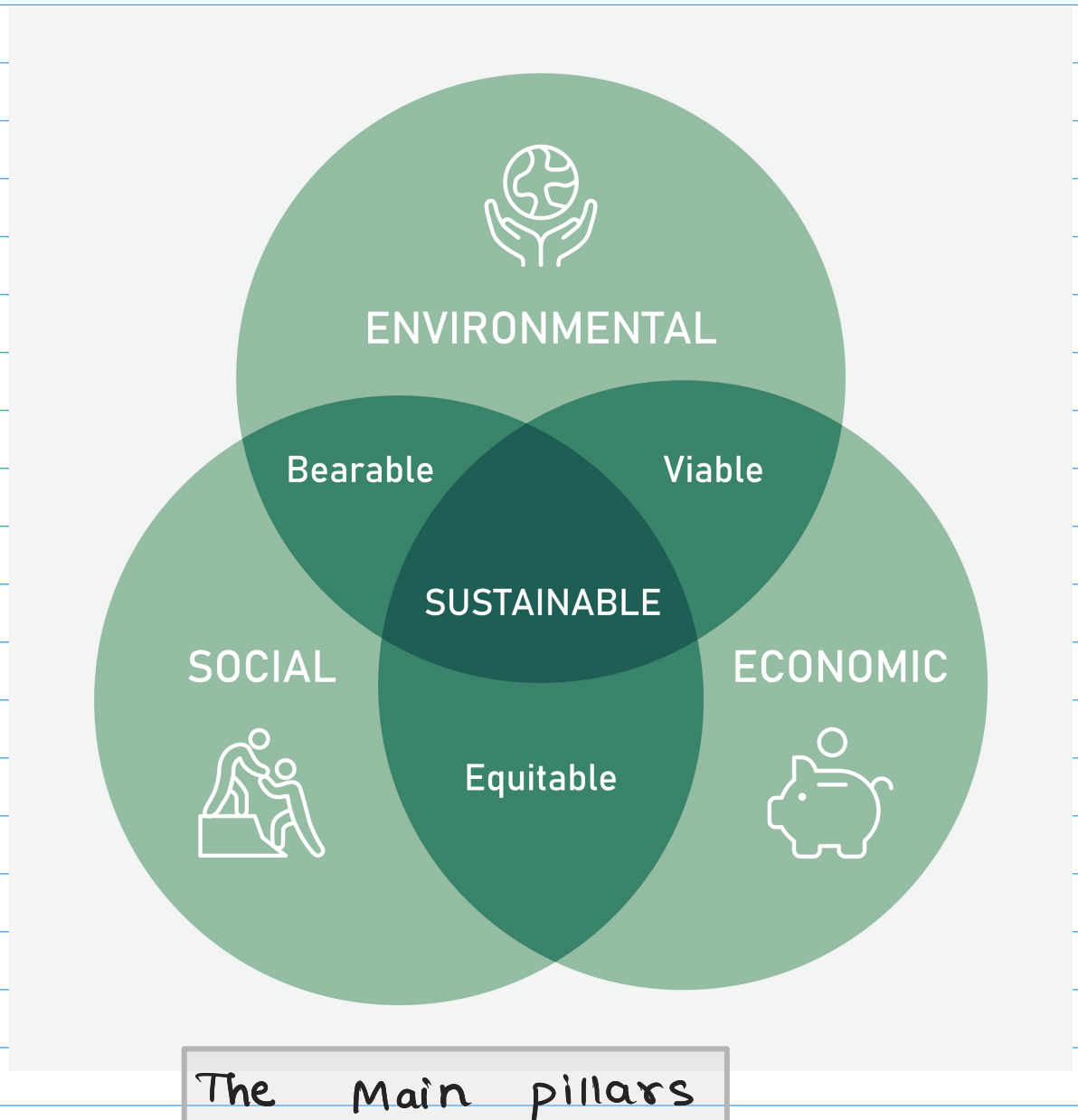
(Ex. - prime minister of Norway)

The Sustainable development definition
was well defined by the report as :

" Development that meets the need of
the present without compromising the
ability of future generations to meet
their own needs "

Break through : It was the first global
report to officially link environmental
health directly with economic development
and social equity.

The pillars : Established that the true
sustainability requires a balance of
three interconnected system.



Economic Growth: Generating long time financial profit and Growth without exhausting resources or exploiting people.

Social: promoting human well-being, fair labour, health and equality.

Environmental : Conserving natural resources and protecting ecosystem to prevent planetary damage.

The overlaps

Bearable : The society can survive comfortably because the environment is healthy, clean and safe to live.

Viable : Business and economics can technically operate in long term because they use clean energy.

Equitable : Wealth and economic opportunities are distributed fairly ensuring basic needs are met without exploitation/ corruption.

The ultimate center

sustainable: The perfect spot where
all the systems are in total balance,
Securing the planet for future generations.

Important years in Sustainability:

1979 - First world climate conference

1987 - The Brundtland Report

1992 - The Rio Earth Summit

1993 - Establishment of UN CSD

1997 - The Kyoto protocol.

2000 - Millennium Development Goals.

2015 - Sustainable Development

Goals and Paris agreement

" CSD - Commission on Sustainable Development "

Green Technology :

- i. Green Technology (Greentech) is a practical engine that turns the concept of sustainability into reality.
- ii. It allows the global economy to expand without burning through the finite resources.
- iii. Economic growth always meant to be more pollution and more resource consumption. Green technology breaks this link by a process called Decoupling.
- iv. Replacing fossil fuels with wind, solar, green hydrogen, next-gen.

nuclear power can scale up power consumption while making the carbon emission zero.

v. Traditional System makes "take-make-waste" model.

Greentech design the waste management well.

vi. Advanced Recycling: chemical technology can break the plastics at their molecular level so they can be reused without losing quality.

vii. Bio-based materials: Engineering biodegradable alternatives to petroleum

based plastics to reduce the plastic pollution.

viii. precision agriculture: using drones, satellites, AI driven sensors allows farmers to exact amount of water and fertilizer required.

ix. This maximize the yield of crops preventing the chemical runoff.

x. Water purification: Advanced membrane technology and solar powered desalination ensure a stable supply of fresh water without draining natural aquifers.

Recycle and examples :

- i. Recycle is the process of collecting and processing materials that would otherwise be thrown as trash and turning them into new products.
- ii. It is the cornerstone of green technology and the circular economy because it prevents waste and saves energy.
- iii. It reduces the need of extracting fresh materials from our planet earth.

Material Category	How it is Recycled	What it Becomes (Examples)
Plastics	Sorted by resin codes (1-7), shredded into flakes, washed, melted, and formed into tiny pellets ready for manufacturing.	* PET bottles (No. 1): Polyester fabric, winter coats, carpets, or new beverage bottles. * HDPE jugs (No. 2): Plastic lumber, playground equipment, and picnic tables.
Metals (Aluminum & Steel)	Melted down in large furnaces. Metal can be recycled indefinitely without losing any of its structural properties.	* Aluminum Cans: Brand new soda cans (often back on shelves in under 60 days). * Scrap Steel: Structural beams, car bodies, and appliances.
Paper & Cardboard	Chopped up and mixed with water to create "pulp." This pulp is washed, bleached, and rolled flat into new sheets.	* Cardboard boxes: Packing materials and paper grocery bags. * Office Paper: Notebook paper, tissues, and paper towels.
Glass	Sorted by color, crushed into small pieces called cullet , melted, and reshaped. Like metal, glass can be recycled infinitely.	* Glass jars: New food jars, glass bottles, and fiberglass insulation.
E-Waste (Electronics)	Manually dismantled to isolate hazardous parts, then crushed to separate valuable trace metals using advanced magnetic and chemical processes.	* Circuit boards & smartphones: Gold, silver, copper, and palladium are extracted and reused in new computer chips and electronics.

Liberal perspective of sustainability:

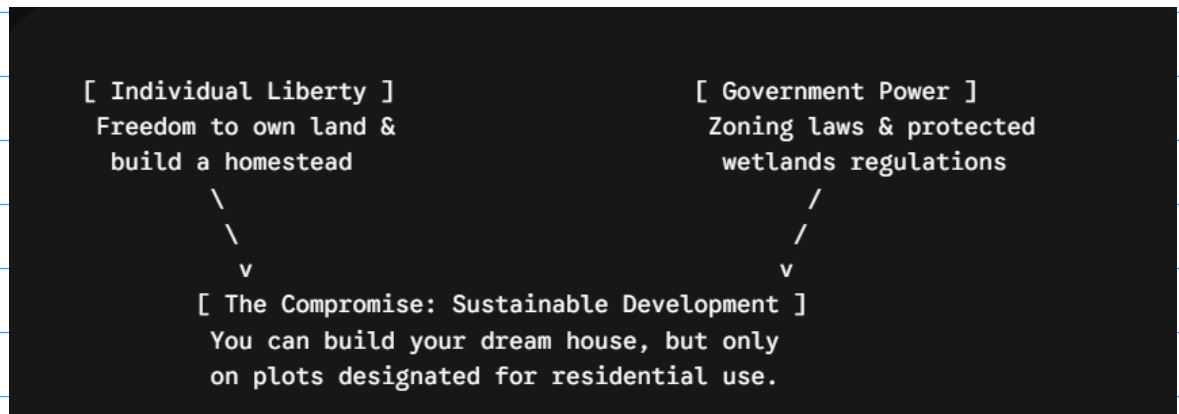
→ It focuses on optimizing existing political and economic frameworks rather than tearing them down.

→ **Regulated Liberty / social contract**

19th century philosopher John Stuart Mill argued that individuals should be completely free to do whatever they want unless their actions cause harm to others.

→ **The Social contract** : when we

live in a society, we implicitly give up our tiny bit of freedom in exchange of protection and benefits of living in a community.



Core pillars of Liberal Sustainability :

1. Green Growth and Market-based solutions :

- This supports Green growth - the idea that economic expansion grow even it is decoupled from environment degradation.
- It makes the market to put taxes on Carbon emission so companies would produce less Carbon.

ii. Techno-Optimism:

- Resource scarcity and climate change are the engineering challenges waiting to be solved.
- Through research and funding, there is a way to clean energy.
(solar, wind, next-generation nuclear) that reduce environmental degradation.

5c's of Sustainable development:

- Sustainability means meet the needs of the present without compromising the ability to meet the needs of future generation by themselves.

→ 5c framework bridges the gap between environmental protection and human progress.

→ It serves as a practical guide on building a sustainable community.

i. Clean (Environment pillar) :

→ This C refers to the ecological health, zero-pollution goals, transition to green energy.

→ It focuses on decarbonization, renewable energy, waste minimizing, keep air, water, soil free from toxins.

ii. Corporate (Economic, Governance pillar):

- This C represents the role of business, industries, organization in sustainability.
- True sustainability must require economic viability but businesses must act ethically.
- It focus on ethics and responsibility of corporate companies.

iii. Community (social pillar):

- This C emphasizes building strong, resilient, supportive communities. Sustainability is fundamentally about people.

→ It focuses on social equity, access to resources, local participation in decision-making.

iv. Culture (Identity and Education pillar):

→ Sustainability development must respect, preserve, integrate local traditions, heritage and values.

→ Culture shapes how people interact with their environment.

→ It focuses on preserving indigenous ecological knowledge and promoting sustainable lifestyle habits.

V. Care (Ethical pillar) :

- This C is the moral compass of the framework. It represents empathy, humanity to protect vulnerable people and non-human life.
- It focuses on biodiversity conservation, animal welfare, prioritize human well being over raw profit.

" These 5Cs are interconnected with each other. IF a Corporate invest in Clean energy project, they demonstrate Care For the Local community Fostering a culture of Responsibility "

